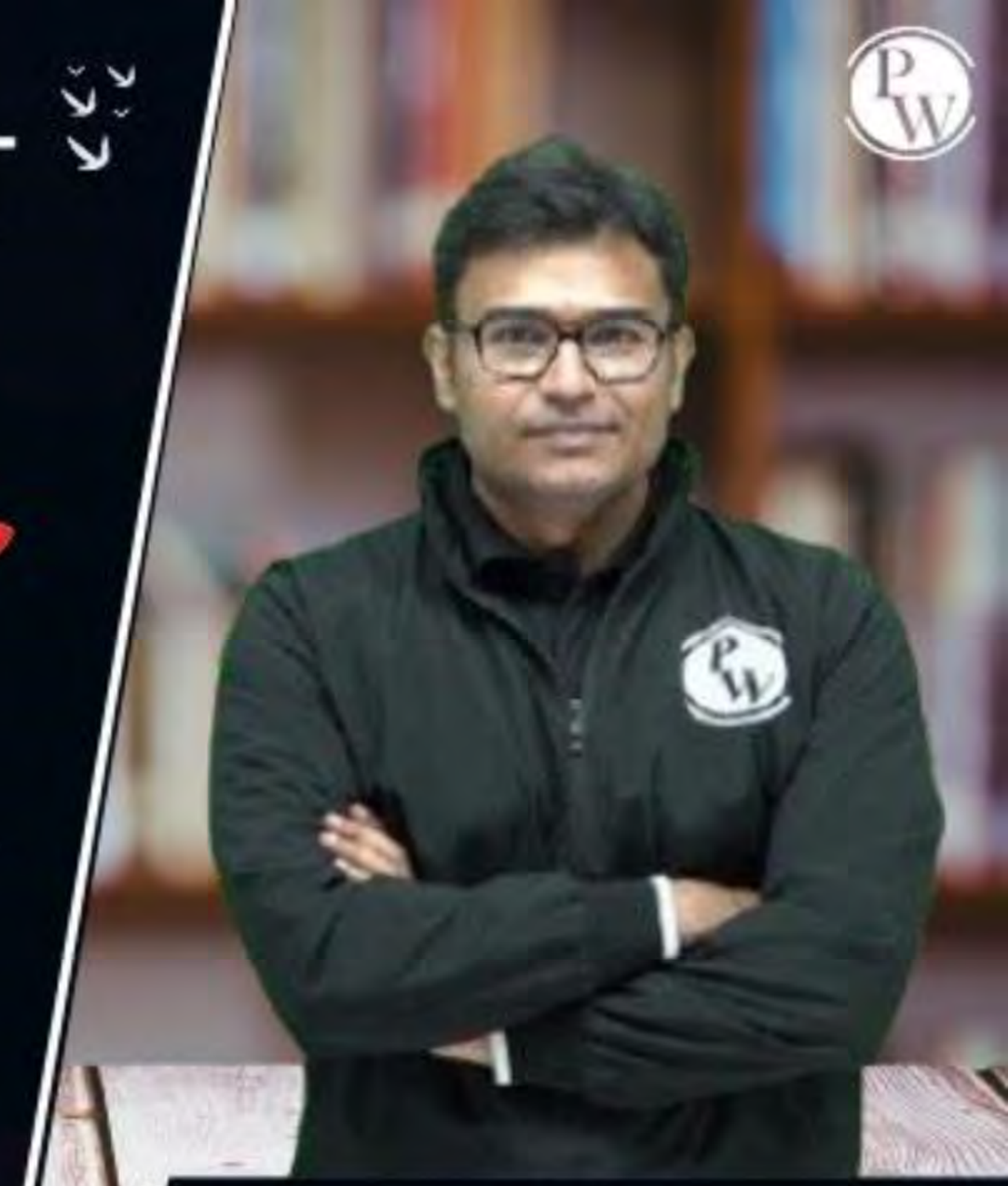
Computer Science & IT

Theory of Computation



CFL

Lecture No. 5



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

CFL

ϵ prod

Unit prod

Topics to be Covered



- Topic
- Topic
- Topic

19 may pm

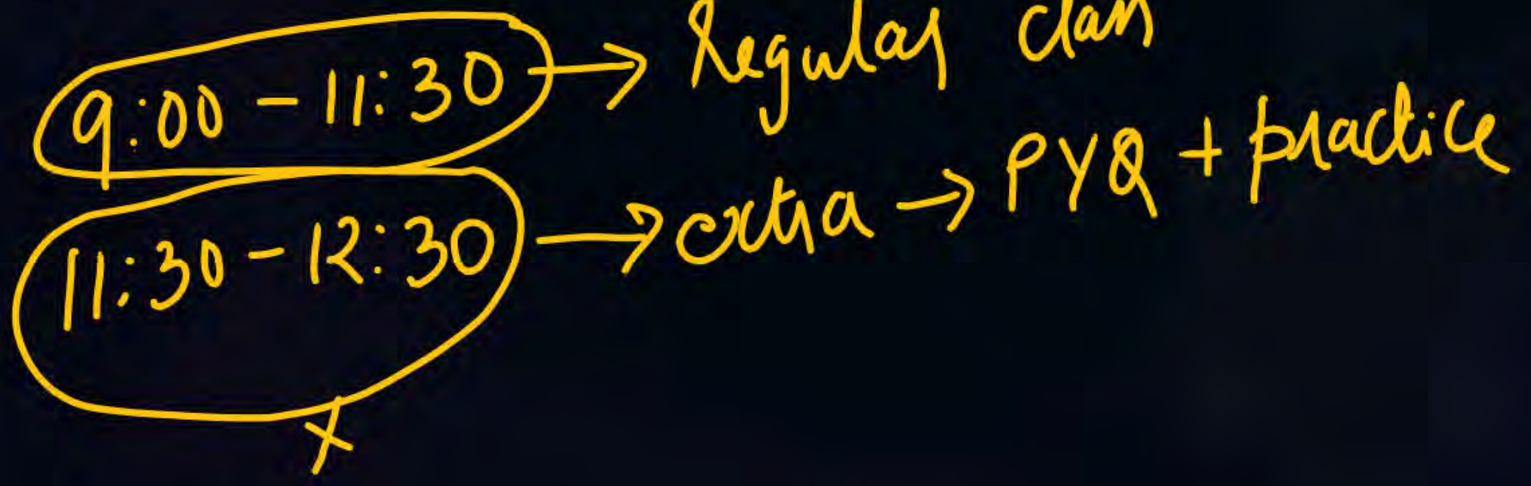
useless production
CNF, GNF
↓
CYK algo

How many

Elimination of useless productions :-

Sometimes a vertex cannot generate any string, then it is useless.

Sometimes a vertex cannot be reached from the start vertex, then it is useless



$S \to AB / a$

$A \to BC / b$

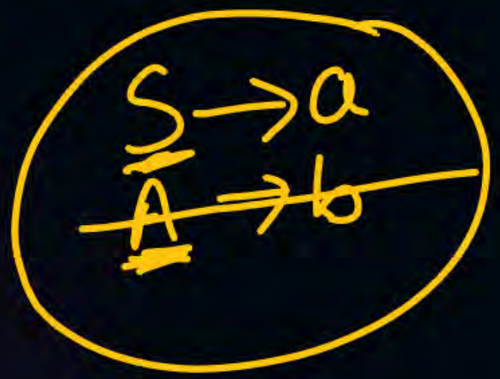
$B \to aB / \epsilon$

$C \to aC / B$

Deriving = $\{a, b, S, A\}$

$V = \{S, A, B, C\}$

$T = \{a, b\}$



Reach

$S \to AB / a$
$A \to BC / b$
$B \to aB / \epsilon$
$C \to aC / B$

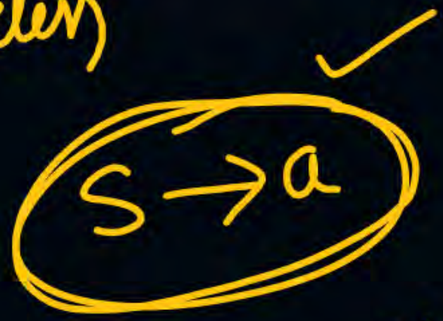
Deriving = $\{a, b, S, A\}$

B, C → useless



$S \to a$
$A \to b$

Reach



Lang = $\{a\}$

$\Sigma = \{a, b\}$

$S \to AB / Aa$

$A \to aAb / bAa / a$

$B \to bbA / aaB / AB$

~~$C \to abCA / aDb$~~

~~$D \to bD / aC$~~

$V = \{S, A, B, C, D\}$

$T = \{a, b\}$



Deriving $= \{a, b, A, B, S,\}$

↓

only a & b

& combination of a, b

C, D → useless

$S \to AB$

$A \to aAb / bAa / a$

$B \to bbA / aaB / AB$

reach A, B

$S \to AB / \cancel{A\epsilon}$

$A \to aAb / bAa / a$

$B \to bbA / aaB / AB$

~~$C \to abCA / aDb$~~

~~$D \to bD / aC$~~

Deriving = {a, b, A, B, S}

S → AB

A → aAb / bAa

B → bbA / aaB / AB





S → ϕ

S → A

ϕ

$S \to ABC / BaB$

$A \to aA / BaC / aaa$

$B \to bBb / a$

$C \to CA / AC$

$S \to BaB$

$A \to aA / aaa$

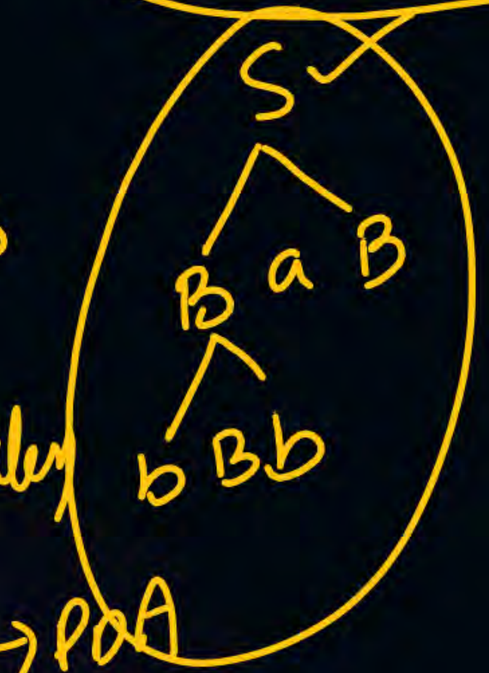
$B \to bBb / a$

Deriving = {a, b, A, B, S}

PDA, CFG, ∈-, ∪, union

Pumping (CYK), CFG → PDA

PDA → CFG



We can represent grammars in many ways

Two popular representations are CNF, GNF

CNF → Chomsky Normal form

GNF → Greibach Normal form

CNF :-

If all productions are of form

$A \rightarrow \underline{BC}$ (A, B, C → vertices) or $A \rightarrow \underline{a}$ (a is terminal)

then the grammar is said to be in CNF

$S \rightarrow AB/a$

$A \rightarrow BC$

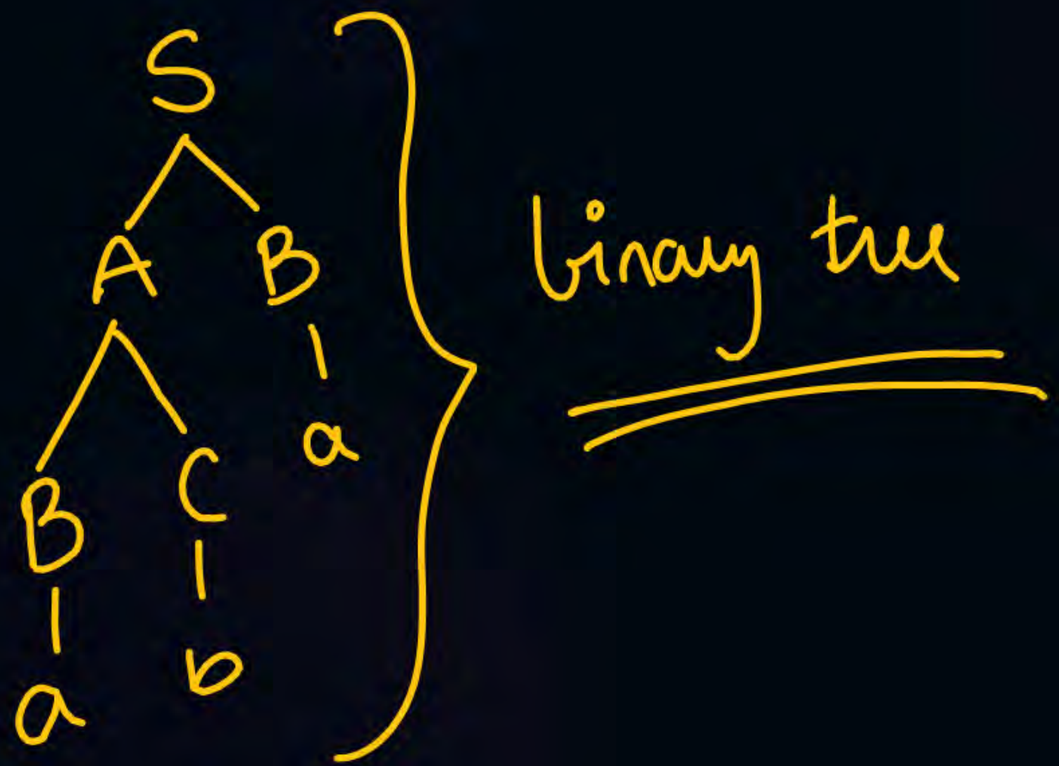
$C \rightarrow CD/b$

S	A	B
S	a	
A	B	C

$S \rightarrow AB$ (1, 2, 3)

$S \rightarrow a$ (1)

length = 3



binary tree

$S \to AB$

$A \to a$

$B \to b$

$S \overset{1}{\Rightarrow} \underline{A}B \overset{2}{\Rightarrow} aB \overset{3}{\Rightarrow} ab$

$2(2)-1$

$S \to SA/b$

$A \to AB/a$

$B \to b$

$S \overset{1}{\Rightarrow} SA \overset{2}{\Rightarrow} \underline{S}AA \overset{3}{\Rightarrow} b\underline{A}A \overset{4}{\Rightarrow} baA \overset{5}{\Rightarrow} \underline{baa}$

3

$2(3)-1$ step

$2|w|-1$

It is easy to apply CYK algorithm

adv of CNF :-

(i) length of the production is restricted

(ii) Derivation tree of parse tree is always binary

(iii) The no of steps required to derive a string of $|w|$ is $2|w|-1$

(iv) It is easy to apply CYK algorithm

C

C Y K

Cocke Younger Kasami

CNF → ε production ✗ → Leyh Shrink

unit production ✗

S → AB → B ε B ⇒

GNF –

A → C

S ⇒ AB ⇒ A D B ⇒ ✓

S ⇒ AB ⇒ CB

Sentencial form

2



at most two children

Derivation :-

Length → increase

& tan

e-p X

U-p X

$S \rightarrow bA/aB$

$A \rightarrow bAA/aS/a$

$B \rightarrow aBB/bS/b$

$S \rightarrow$ equal no a's b's

$S \rightarrow bA/aB$

$A \rightarrow a/aS/bAA$

$B \rightarrow b/bS/aBB$

$S \rightarrow$ equal no of a's b's

$S \to bA / aB$

$A \to a / aS / bAA$

$B \to b / bS / aBB$

$S \to N_bA / N_aB$

$A \to a / N_aS / N_bAA / X$

$B \to b / N_bS / N_aBB / Y$

$N_a \to a$

$N_b \to b$

$X \to AA$

$Y \to BB$

$A \to BC$

$A \to a$

CYK

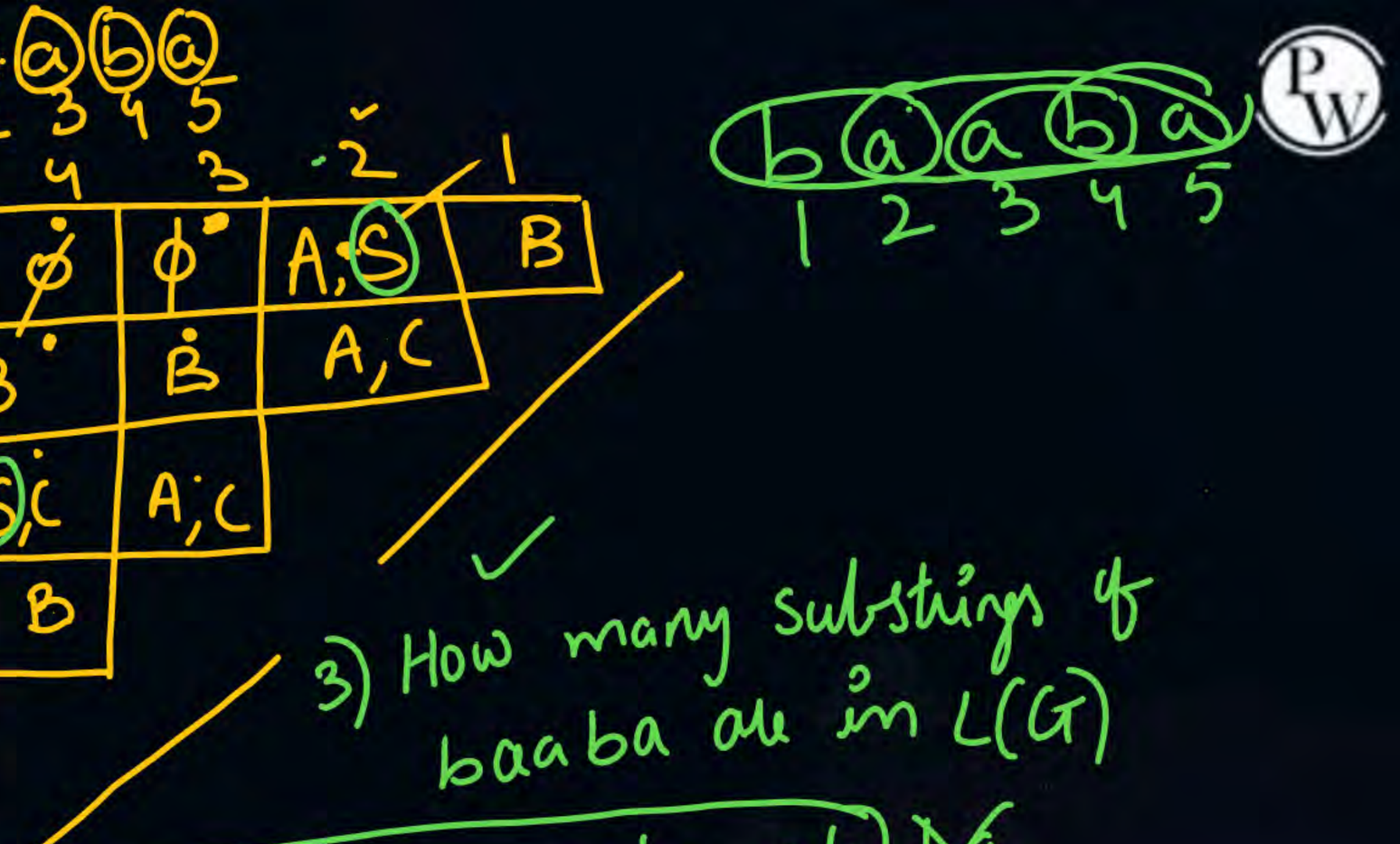
G:

$S \to AB/BC$

$A \to BA/a$

$B \to CC/b$

$C \to AB/a$



baaba $\in L(G)$

$w \in L(G)$

Substrings of $w \in L(G)$

3) How many substrings of baaba are in L(G)

baaba, ba, aaba, ab ~~ba~~

4

a a b b b

$S \rightarrow AB$
$A \rightarrow BB/a$
$B \rightarrow AB/b$

$S \to AB$
$A \to BB/a$
$B \to AB/b$

a a b b b
1 2 3 4 5

	5	4	3	2	1
1	S,B	A	S B	∅	A
2	S B	A	S B	A	
3	S B	A	B		
4	A	B			
5	B				

what are the Substrings generated by 'A'.

aabb, abb, bb, bb, a

aabbb $\stackrel{?}{\in}$ L(G)

what variables derive aabbb?

S, B

what are the substring of aabbb that belong to L(G)

aabbb, aab, abbb, ab, bbb

GNF:-

→ If all productions are of form $A \to a\alpha$ $a \in \Sigma$, $\alpha \in V^*$, then it is called a GNF

$A \to aB$

$B \to bC$

$C \to c$

$A \xRightarrow{1} aB \xRightarrow{2} abC \xRightarrow{3} (abc)$

$|w| \Rightarrow |w|$ steps

$A \to aB$

$B \to bBC / a$

$C \to aBAC / b$

Advi

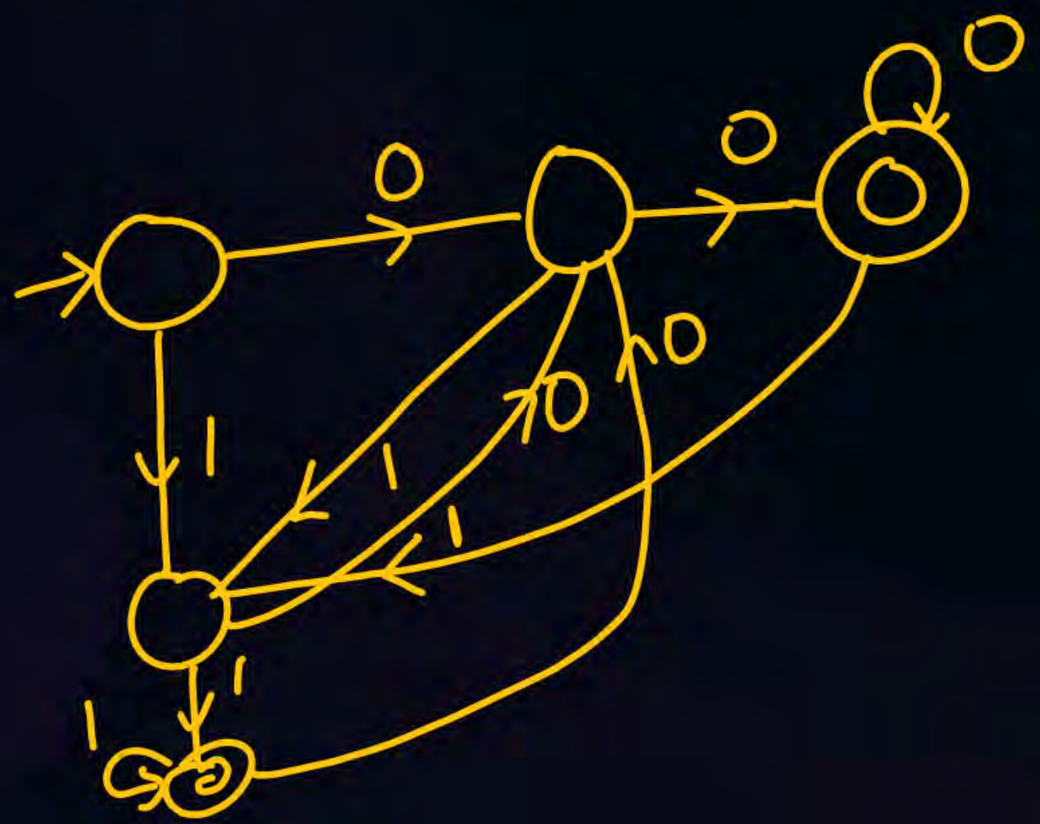
→ The no of steps required to generate a string of length $|w|$ = $|w|$

→ GNF is used to convert CFG to PDA

5 min Break

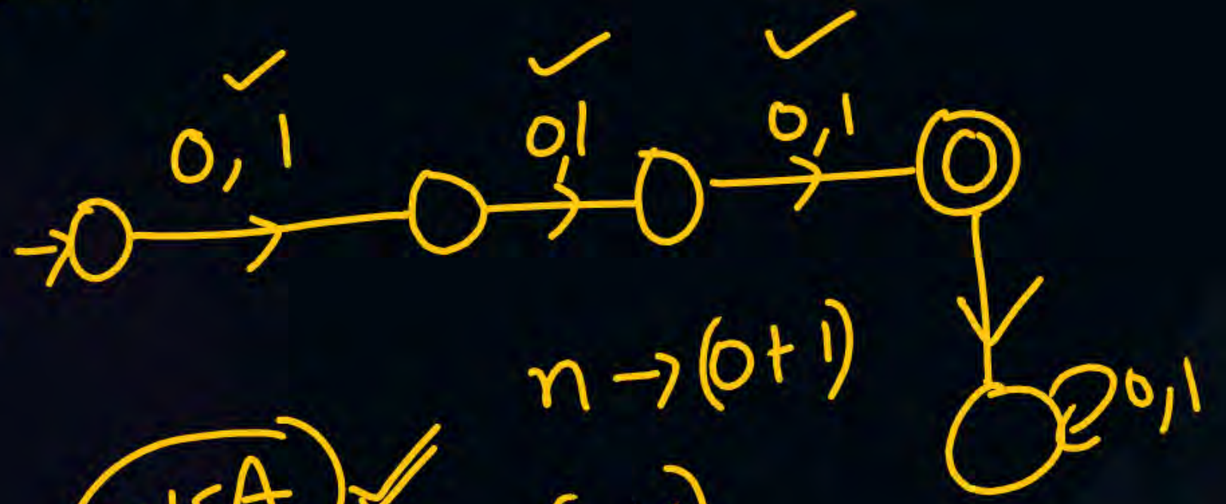
Let L be the set of all binary strings whose last two symbols are same. How many states are in min DFA

a) 2 b) 5 c) 8 d) 3



Consider the regular expression $(0+1)(0+1)$ ---- n times
The minimum state finite automata that recognizes this language contains

a) n states
b) (n+1) states
c) (n+2) states
d) none

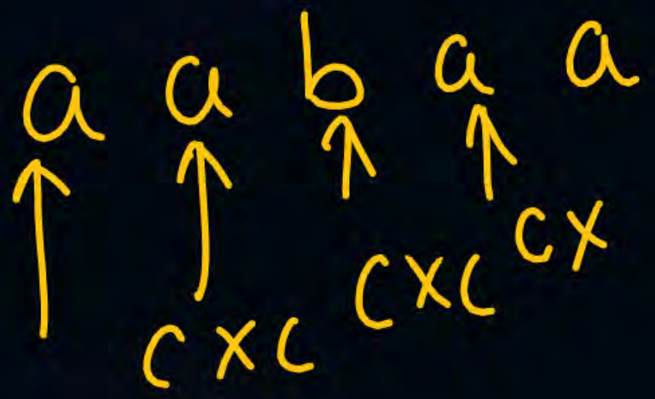


NFA ✓

$n \to (0+1)$

$(n+1)$

min DFA $\to$ (n+2) - dead


a a b a a
cxc cxc cx

Let S and T be languages over $\Sigma = \{a, b\}$ represented by regular expressions $(a+b^*)^*$ and $(a+b)^*$ respectively. which of the following is true

a) $S \subset t$ b) $T \subset S$ c) $S = t$ d) $S \cap T = \phi$

$(a+b)^* = (a^*+b)^* = (a+b^*)^* = (a^*b^*)^*$

ϵ, b, bb

Let L denote the language generated by the grammar

$S \to 0S0/00$. which of the following is true

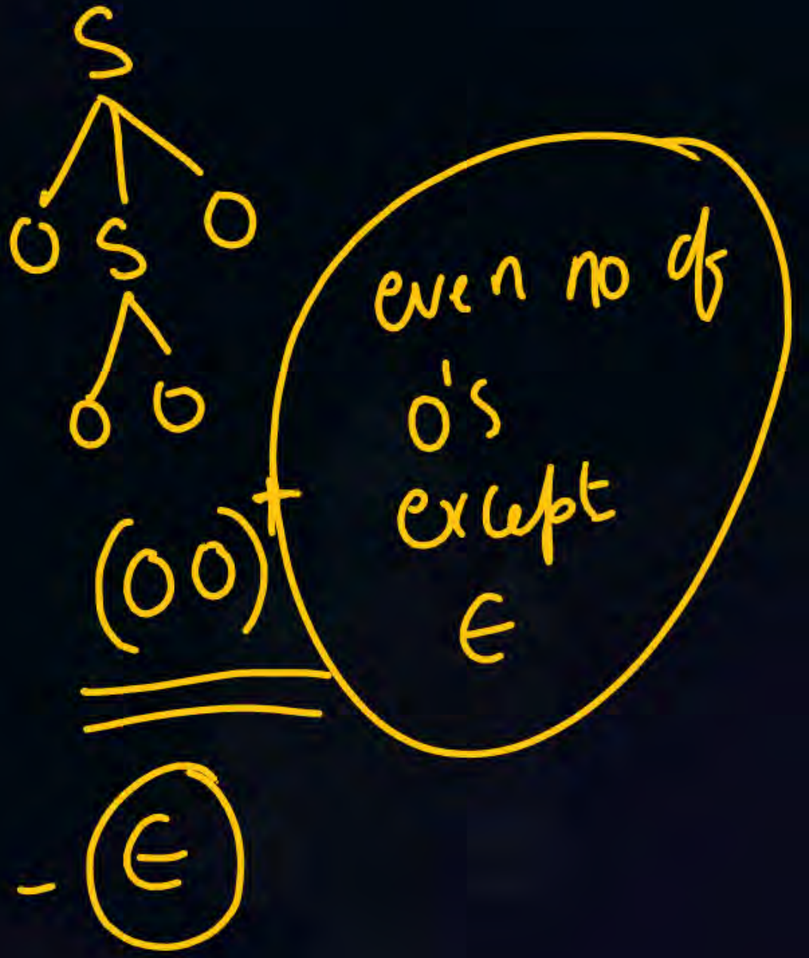
a) $L = 0^+$

b) L is regular but not 0^+

c) L is CFL but not regular

d) L is not CFL

CN → 150 hour



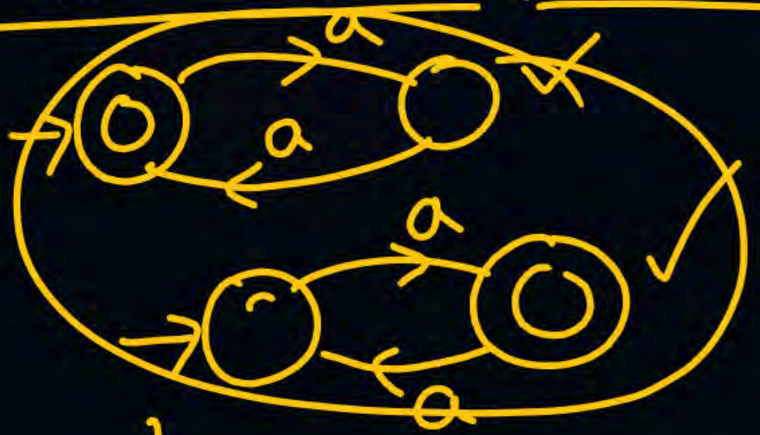
What can be said about a regular language 'L' over {a} whose minimal finite automata has two states

a) Can be $\{a^n / n \text{ is odd}\}$

b) Can be $\{a^n / n \text{ is even}\}$

c) Can be $\{a^n / n \geq 0\}$ ✓

d) Either L can be $\{a^n / n \text{ is odd}\}$ & L can be $\{a^n / n \text{ is even}\}$





Consider the following statements

S_1: $\{0^{2n} / n \geq 1\}$ is regular

S_2: $\{0^m 1^n 0^{m+n} / m, n \geq 1\}$ is regular

a) only S_1 is true
b) only S_2 is true
c) Both S_1, S_2 are true
d) None of S_1, S_2 are true

DCFL ?

Given an arbitrary NFA with N states, the max number of states in an equivalent minimized DFA is atleast

a) N^2 b) 2^N c) $2N$ d) $N!$

2nd Sym from RHS is a — NFA (3) — DFA (4)

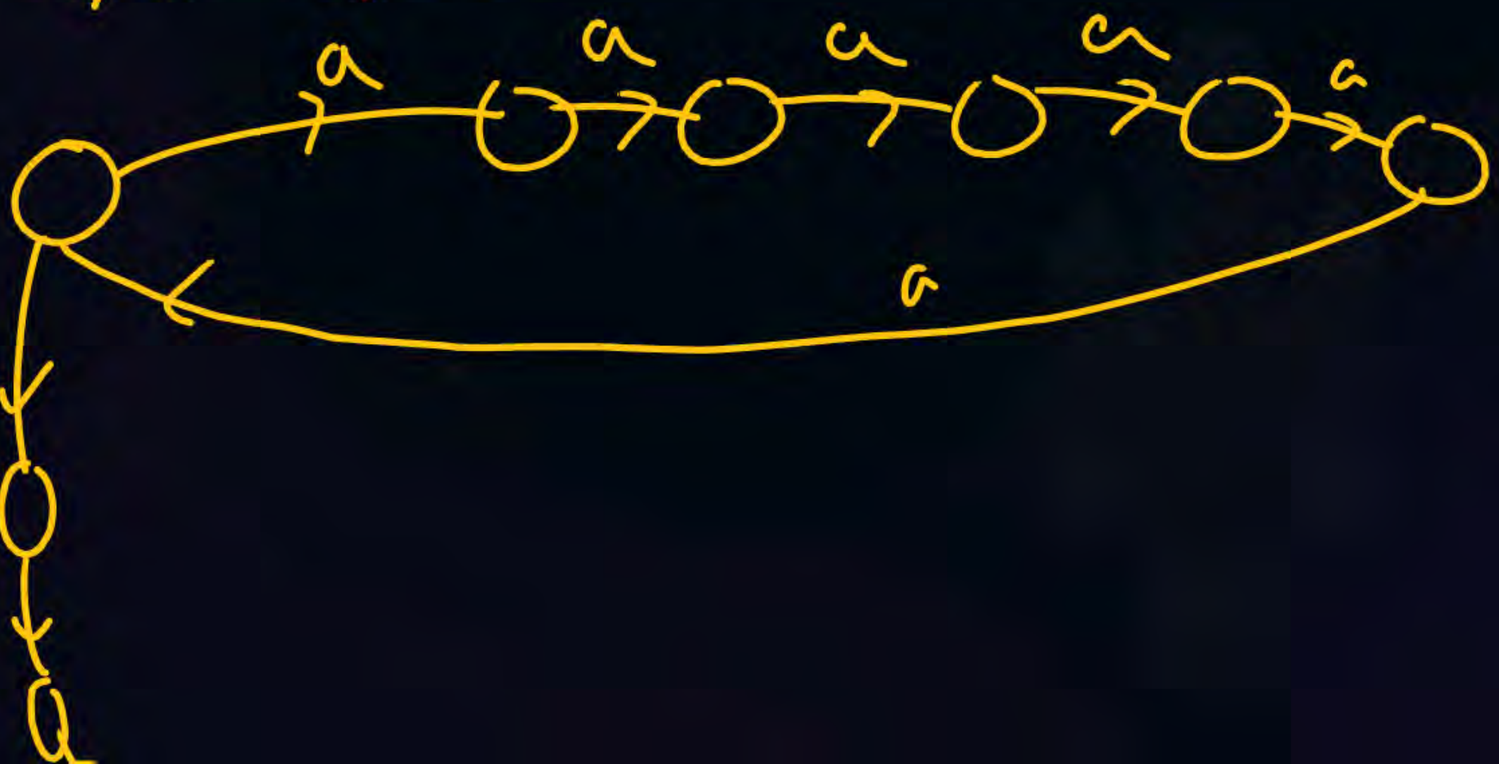
3rd Sym from RHS is a — NFA (4) — DFA (8)

$O(2^n)$

Consider a DFA over $\Sigma = \{a, b\}$ accepting all strings which have num of a's divisible by 6 and no of bs divisible by 8. minimum no of states in DFA?

a) 8 b) 14 c) 15 d) 48

Std $6 \times 8 = 48$



Consider the following languages

$L_1 = \{ww / w \in (a,b)^*\} \longrightarrow$ CSL

$L_2 = \{ww^R / w \in \{a,b\}^*\} \longrightarrow$ CFL DCFL ? X

$L_3 = \{0^{2^i} / i \geq 1\} \longrightarrow$ Reg

$L_4 = \{0^{i^2} / i \geq 1\} \longrightarrow$ CSL

Which of the languages are Reg?

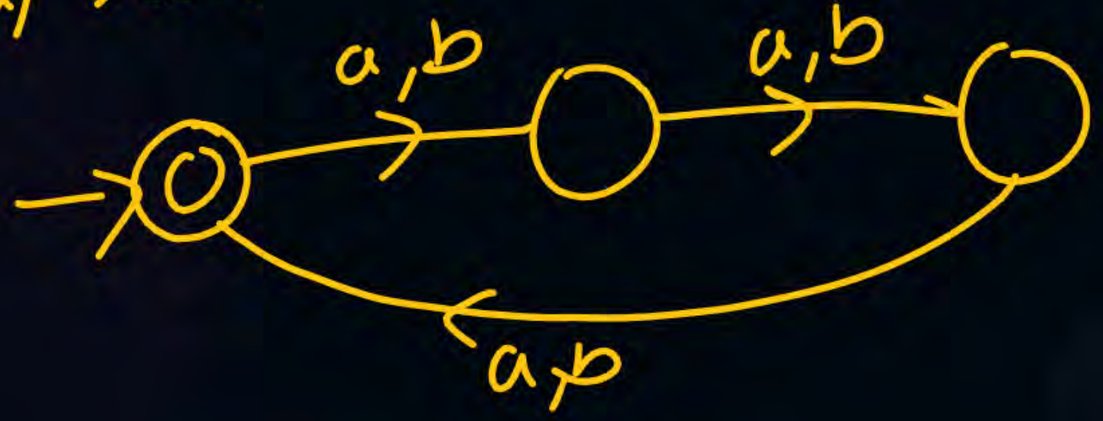
a) L_1, L_2 b) L_2, L_3, L_4

c) L_3, L_4 d) L_3

The smallest finite automata which accepts the language $L = \{ x / \text{length of } x \text{ is divisible } 3 \}$ has

a) 2 states b) 3 states

c) 4 states d) 5 states



THANK - YOU